Experiment 7

EDA – Quantitative and Qualitative Analysis.

Program:

```
import seaborn as sns
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
tips = sns.load_dataset('tips')
print(tips.head())
sns.displot(tips.total_bill, kde=True)
sns.displot(tips.total_bill, kde=False)
sns.jointplot(x=tips.tip, y=tips.total_bill)
sns.jointplot(x=tips.tip, y=tips.total_bill, kind="reg")
sns.jointplot(x=tips.tip, y=tips.total_bill, kind="hex")
sns.pairplot(tips)
print(tips.time.value_counts())
sns.pairplot(tips, hue='time')
sns.pairplot(tips, hue='day')
sns.heatmap(tips.corr(numeric_only=True), annot=True)
sns.boxplot(x=tips.total_bill)
sns.boxplot(x=tips.tip)
sns.countplot(x=tips.day)
sns.countplot(x=tips.sex)
tips.sex.value_counts().plot(kind='pie')
tips.sex.value_counts().plot(kind='bar')
sns.countplot(x=tips[tips.time=='Dinner']['day'])
print("All plots generated successfully.")
```

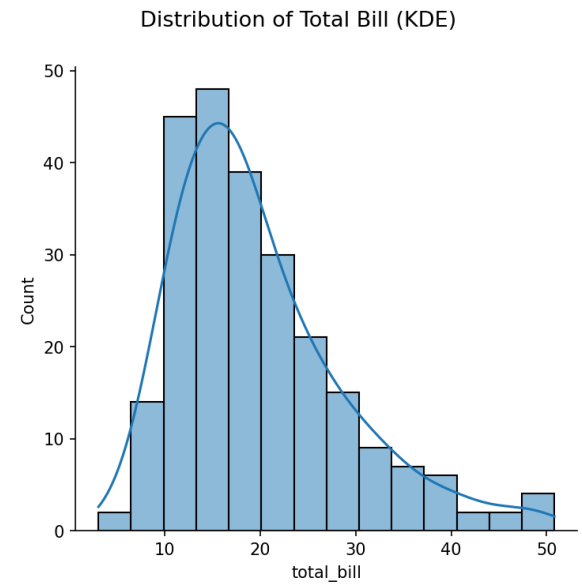
Output

```
   total_bill   tip     sex smoker  day    time  size
0       16.99  1.01  Female     No  Sun  Dinner     2
1       10.34  1.66    Male     No  Sun  Dinner     3
2       21.01  3.50    Male     No  Sun  Dinner     3
3       23.68  3.31    Male     No  Sun  Dinner     2
4       24.59  3.61  Female     No  Sun  Dinner     4

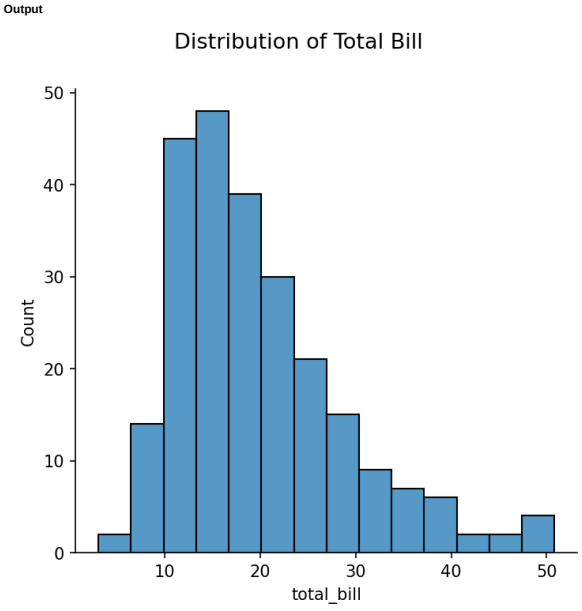
time
time
Dinner    176
Lunch      68
Name: count, dtype: int64

All plots generated successfully.
```

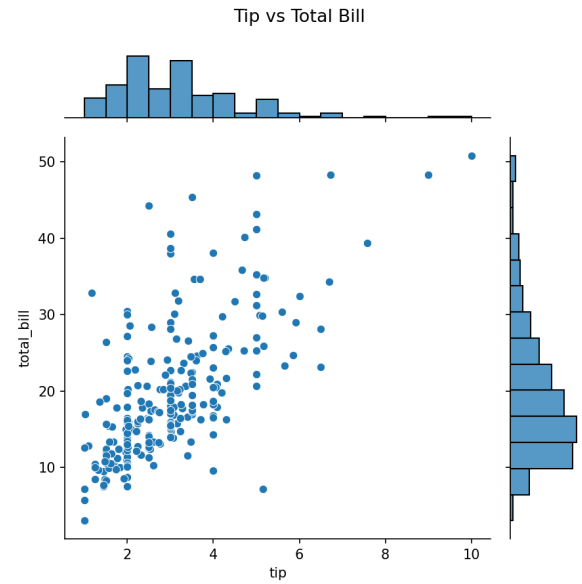
Output



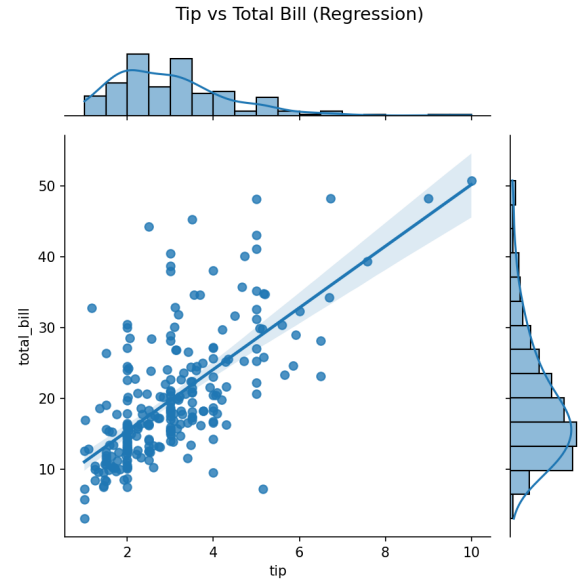
Output



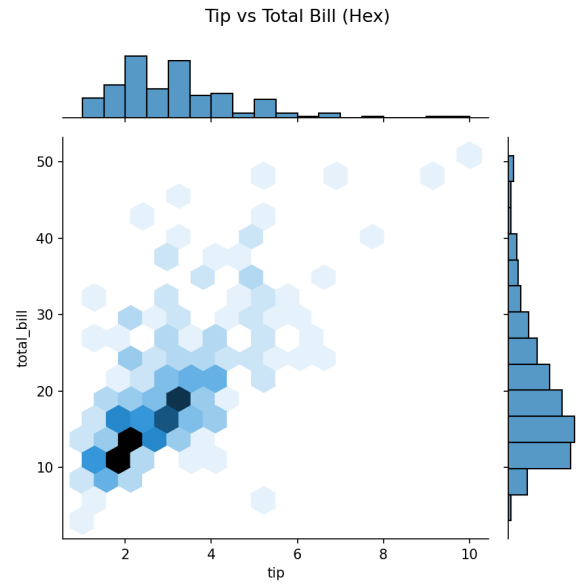
Output



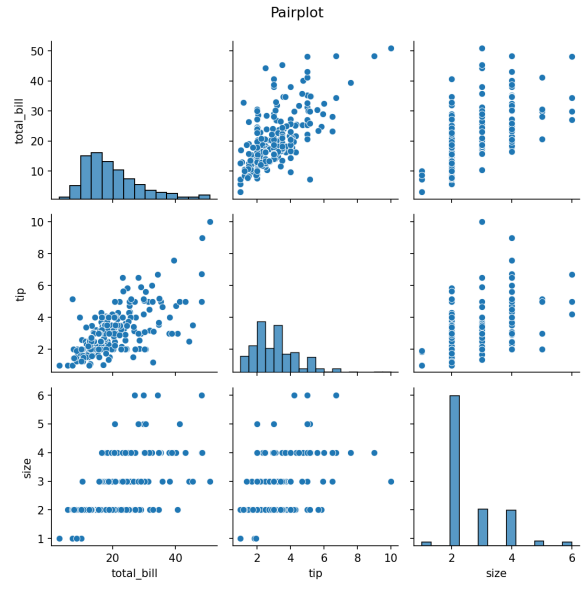
Output



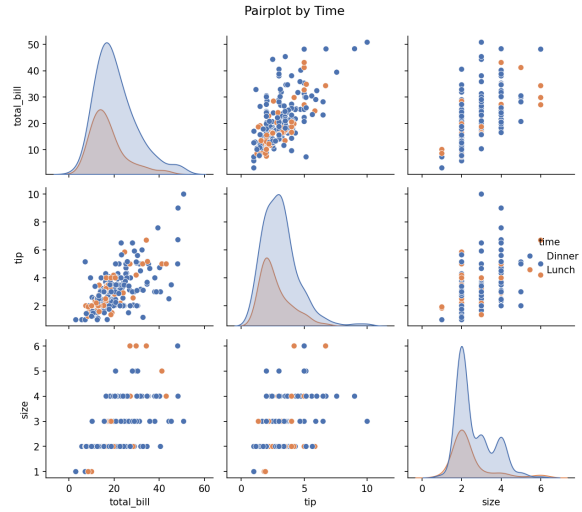
Output



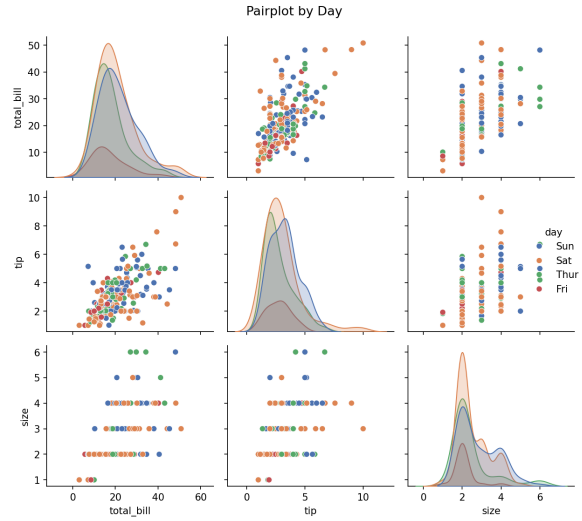
Output



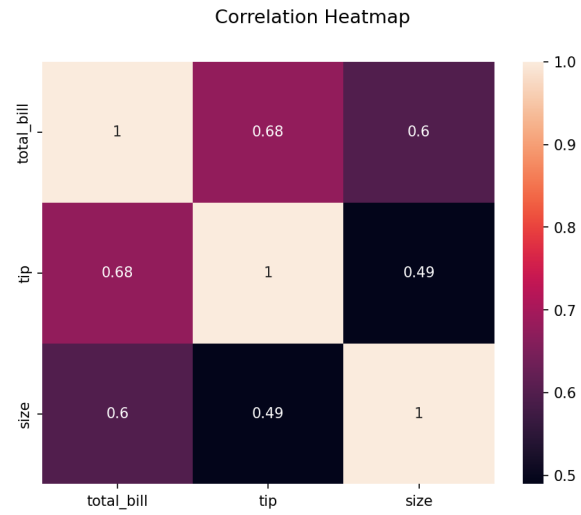
Output



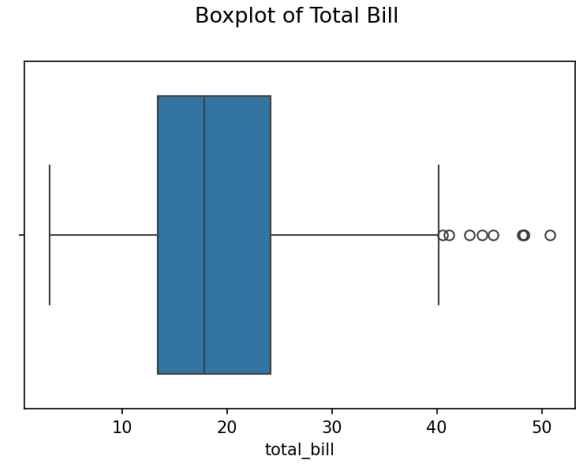
Output



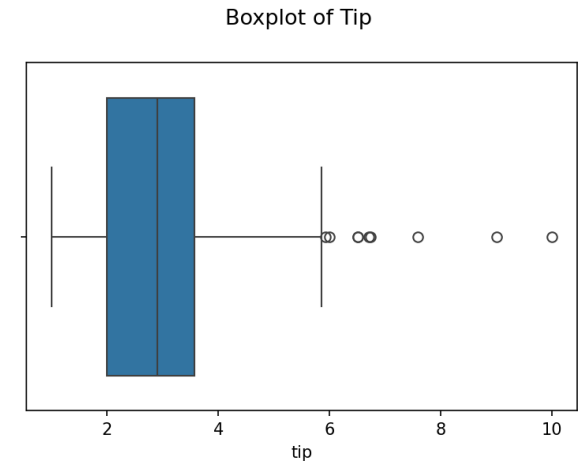
Output



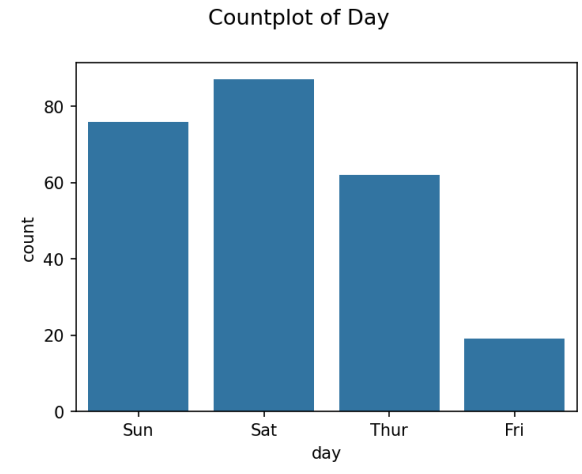
Output



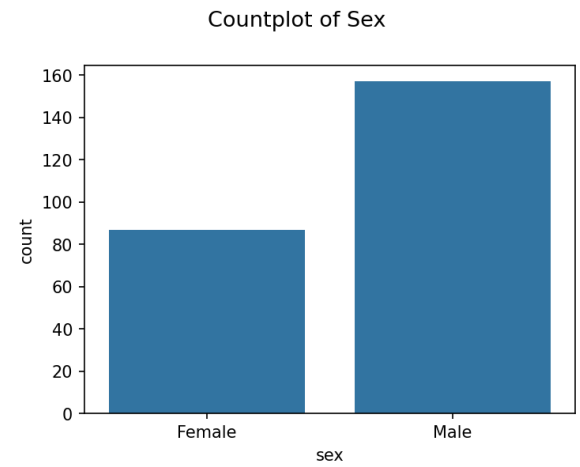
Output



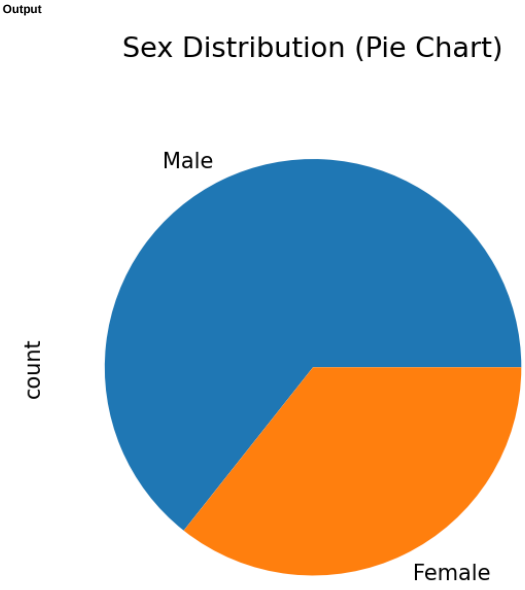
Output



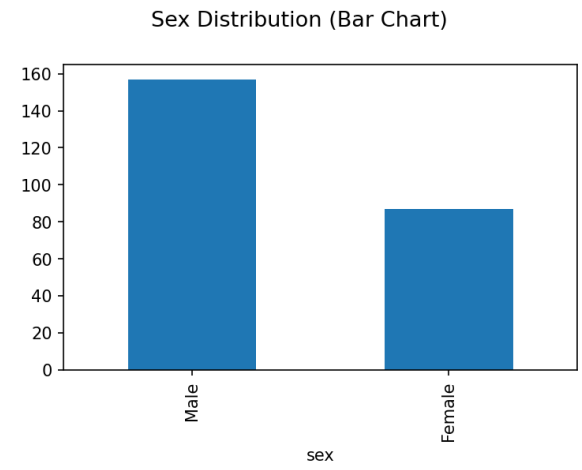
Output



Output



Output



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